Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping **React**, **TypeScript**, and **Node.js** features over relational SQL, building production LLM workflows with output validation, and treating **testing**, authorization, and access control as first-class requirements.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized multi-tenant schemas on Azure SQL with **row-level security** isolating tenant data for 500+ users, and tuned indexed queries to cut dashboard latency from 1.2s to 350ms.
- Restructured asynchronous **LLM orchestration** with FastAPI workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput, and shipped PostgreSQL integrations through CI/CD at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped a full-stack **React** and Flask grading platform to production on AWS, owning upload, rubric, grading, and history flows end to end for 150+ instructors and 10K+ daily requests.
- Built LLM and OCR workflows across GPT-4o, Claude, and Qwen-VL with vision fallbacks, per-call cost logging, and **output validation**, holding 90%+ grading consistency across 10K+ submissions and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated 20+ **unit and integration test** suites with PyTest and Mockito in nightly Jenkins pipelines, treating verification as part of every release across Windows, Linux, and macOS builds.
- Optimized large-scale API integrations with pagination, field filtering, indexing, and **caching**, improving response times 30% for low-bandwidth users and cutting technical debt 20% through object-oriented module refactoring.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **React, TypeScript, Node.js, and Express** platform with 8 microservices including JWT authentication and **role-based access control**, syncing 2M+ records across locations through real-time WebSocket updates.
- Modeled relational data in PostgreSQL and Azure SQL with encryption and access controls, and gated releases on **Jest**, Selenium, and Snyk checks in Jenkins CI/CD, reaching 95%+ successful build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built a **React and TypeScript** web dashboard and Flask booking APIs with time-slot selection and Firebase transactional locking to prevent double booking, keeping responses under 200ms across 50K+ records.
- Designed the end-to-end architecture and **presented it to the client** before implementation, then automated refunds and disputes through idempotent payment webhooks, cutting manual payment handling about 70%.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, HTML, CSS
Full Stack: React, Next.js, Node.js, Express.js, REST APIs, Flask, FastAPI, WebSockets
Data & Security: PostgreSQL, Azure SQL, MySQL, Schema Design, Row-Level Security, JWT, Role-Based Access Control, Encryption
AI & Testing: LLM APIs, OCR, Prompt Engineering, Jest, Selenium, PyTest, Unit Testing, Integration Testing
Cloud & Practices: AWS, Azure, Docker, GitHub Actions, CI/CD, Git, Code Reviews, Agile, Sentry

Projects

Imaging Appointment Scheduling Platform | *TypeScript, React, Node.js, Express, PostgreSQL, Jest*

- Designed a **TypeScript** scheduling app where patients and care coordinators book imaging exams across provider sites, with PostgreSQL migrations, role-based authorization, and **audit logging** of every protected health record access.

Referral Intake Agent with Evals | *TypeScript, Node.js, LLM APIs, OCR, PostgreSQL, FHIR*

- Designed an **LLM agent** that extracts exam orders from scanned referral PDFs via OCR into FHIR-shaped records, with schema guardrails, confidence-based human review, and an **eval suite** measuring field-level extraction accuracy.

HMIS – Hospital Management Information System Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*

- Won Smart India Hackathon building a **healthcare** platform with consent-based patient history access through ABDM APIs, appointment booking and rescheduling, Twilio reminders, and Sentry monitoring, cutting administrative workload 30%.